

Tumour	Gene_Symbol	HLA_allele	Mut_peptide	Norm_MHCrank	Mut_MHCrank_EL
WD06	ACKR3	HLA-A02:01	LLYHVAVLL	12.6796	0.3276
WD06	ADAM22	HLA-A02:01	FLYVTSVMM	7.1905	0.4538
WD06	ATXN2L	HLA-A24:02	LYPFPPrIL	11.4784	0.1464
WD06	BBS4	HLA-A24:02	ACAFPEsPPL	3.1733	0.44
WD06	C1orf186	HLA-A24:02	WYFVNPALF	7.2642	0.0198
WD06	C1orf186	HLA-A24:02	FWYFVNPAL	9.9507	0.1198
WD06	C1orf186	HLA-A24:02	SFWYFVNPA	17.8696	0.1396
WD06	C1orf186	HLA-A24:02	YFVNPALF	23.2727	0.1874
WD06	CCDC73	HLA-A02:01	KLEQNVWEA	3.9909	0.276
WD06	CNBD1	HLA-A24:02	IWSEEFKNF	9.5501	0.0417
WD06	DRC1	HLA-A02:01	KILEAFVMV	11.6109	0.0767
WD06	FERMT1	HLA-A24:02	QYHISKLSLF	11.1005	0.1759
WD06	FERMT1	HLA-A24:02	YHISKLSLF	22.6974	0.2916
WD06	FETUB	HLA-A24:02	KYNNENTF	28.0294	0.4095
WD06	FKBP5	HLA-A24:02	EYTKAVECF	13.8536	0.2615
WD06	GALNT6	HLA-A24:02	VYSVLHTI	10.6378	0.2864
WD06	GALNT6	HLA-A02:01	TVYSVLHTI	2.0866	0.4802
WD06	GMCL1	HLA-A24:02	TGFNFGFDLI	2.7631	0.4953
WD06	HJURP	HLA-A24:02	KLPSSPLGF	12.4178	0.2639
WD06	HMGXB4	HLA-A24:02	KKYPPTTMLL	3.8303	0.0932
WD06	HMGXB4	HLA-A24:02	QKKYPPTTMI	10.7483	0.3731
WD06	IRS4	HLA-A02:01	MMFPRVSPL	5.6535	0.306
WD06	KALRN	HLA-A24:02	GYLDYNHEE\	7.2806	0.1817
WD06	KDEL2	HLA-A24:02	SFVADSHQW	11.7354	0.4588
WD06	KIAA1217	HLA-A24:02	YYNPSMQAE	23.9242	0.4279
WD06	KIAA1549L	HLA-A24:02	MYARTGHTN	3.8282	0.1307
WD06	KIAA1549L	HLA-A24:02	YMYARTGHT	8.9206	0.3824
WD06	MITF	HLA-A24:02	TFDLTDGTITI	2.6104	0.342
WD06	MUC16	HLA-A24:02	TYTMGETSVI	16.25	0.084
WD06	MUC16	HLA-A24:02	SVFISTSDF	5.2911	0.4977
WD06	MUC17	HLA-A02:01	TLVDSKTQV	22.7453	0.2551
WD06	OOSP2	HLA-A24:02	VYEFIYLV	5.8408	0.2909
WD06	OR2T11	HLA-A24:02	TYVLPQSFHI	7.9594	0.2206
WD06	OR4D10	HLA-A24:02	IFLLPLPF	2.7734	0.32
WD06	OR4D2	HLA-A02:01	FSMDKLVS	2.3391	0.1516
WD06	OR51B2	HLA-A24:02	QYGIIRLLF	7.5499	0.0316
WD06	OR51B2	HLA-A24:02	IQYGIIRLLF	16.7366	0.311
WD06	PCDHA12	HLA-A24:02	IFYGIKMIL	3.6844	0.4301
WD06	PCDHB7	HLA-A02:01	GIWAGAEPL	3.2677	0.4203
WD06	PCDHGB3	HLA-A24:02	YLSPDPHFF	17.3207	0.1828
WD06	PIEZO2	HLA-A02:01	GLWKSSPLL	3.353	0.2501
WD06	PIEZO2	HLA-A02:01	GLWKSSPLL	2.0226	0.0627
WD06	PLK4	HLA-A24:02	QYLSSILLMF	6.2008	0.1075
WD06	PLK4	HLA-A24:02	QYLSSILLM	20.7739	0.1415
WD06	PTPN4	HLA-A24:02	YYQKMPVIV	15.976	0.1334

WD06	RAD51B	HLA-A24:02	AFLSTTLF	33.7059	0.4412
WD06	RFPL1	HLA-A24:02	TYGRHYWEV	26.1628	0.3236
WD06	RPS6KA5	HLA-A02:01	YINGGELFI	2.8827	0.2101
WD06	SIGLEC12	HLA-A24:02	SYPQNGWT/	7.2538	0.0156
WD06	SIGLEC12	HLA-A24:02	YWFRAGDH\	15.8048	0.4129
WD06	SIGLEC12	HLA-A24:02	FSYPQNGW1	20.5478	0.4933
WD06	SIGLEC12	HLA-A24:02	AFDPVHGYW	2.5995	0.1651
WD06	SLC17A8	HLA-A24:02	VFGAAIFLI	5.0871	0.2267
WD06	SLC2A10	HLA-A24:02	AFLQSLSLLF	2.7809	0.3316
WD06	SMARCC2	HLA-A24:02	KYIQAEPPi	9.6063	0.2968
WD06	SNX13	HLA-A24:02	SFIKSSERW	16.433	0.214
WD06	SNX13	HLA-A24:02	GYRPTAARSI	6.3408	0.3424
WD06	ST3GAL1	HLA-A02:01	LMMNKAPT/	17.1889	0.359
WD06	TCAF2	HLA-A02:01	FTQLFDEYQT	2.3457	0.2998
WD06	TIGD4	HLA-A24:02	KFIELAFF	4.551	0.3549
WD06	UBR4	HLA-A24:02	RYNLDDYRI	6.6729	0.1617
WD06	WHAMM	HLA-A24:02	PYASINEHI	3.8361	0.0897
WD06	WHAMM	HLA-A24:02	PYASINEHIL	9.1177	0.3221
WD06	ZNF333	HLA-A02:01	YTFTEILSI	5.3487	0.2832
WD06	ZNF880	HLA-A24:02	KYLDPAQRTL	7.3415	0.1605
WD06	ZYX	HLA-A24:02	ASFPAPAPKF	3.2478	0.2079
WD13	ASB10	HLA-A02:01	TLLHVAARL	21.4885	0.3753
WD13	CLEC7A	HLA-A24:02	YLFSMSLNF	25.7347	0.4596
WD13	DACH1	HLA-A02:01	LLSHPLNHL	4.6679	0.3821
WD13	DNAH2	HLA-A02:01	KQVESILHL	8.7578	0.3439
WD13	EBF2	HLA-A02:01	RLDPSEATL	5.5451	0.1072
WD13	EOGT	HLA-A24:02	GYQNTGLFR	31.1739	0.4757
WD13	FSD1L	HLA-A02:01	NLSNILSEL	10.2834	0.2562
WD13	GLI3	HLA-A02:01	SLFSPPHPYI	11.1825	0.1557
WD13	GOLGA4	HLA-A02:01	KMDEQIKTI	2.3733	0.1049
WD13	HAUS8	HLA-A02:01	KMMESQTLL	3.2742	0.3986
WD13	HTR1E	HLA-A02:01	KVADFLTWL	2.3472	0.1248
WD13	IPO4	HLA-A24:02	FFGPPPLCW	6.9861	0.1652
WD13	IPO4	HLA-A24:02	SFFGPPPLCV	7.7875	0.2002
WD13	KRTAP19	HLA-A24:02	YYGGLGYGY	2.7907	0.4163
WD13	LIMK2	HLA-A24:02	IWYRTVNEI	7.2423	0.2321
WD13	MAP3K9	HLA-A02:01	RLPDPNVVFI	5.4464	0.1883
WD13	MAST2	HLA-A02:01	FLSRDPRSL	23.022	0.4971
WD13	MLLT6	HLA-A02:01	YLPDTPICL	2.2018	0.0903
WD13	MYH14	HLA-A02:01	FQAGVLAQL	5.0063	0.2115
WD13	NOD2	HLA-A02:01	GLIWLIRSL	7.7211	0.4771
WD13	NPAS2	HLA-A02:01	LLVPGLSQA	17.6407	0.2142
WD13	OR5D18	HLA-A24:02	SYAWGVSCF	11.8193	0.0494
WD13	OR8U1	HLA-A24:02	LYPLMHVLL	5.3451	0.1363
WD13	OR8U1	HLA-A02:01	ALVDFCYSSV	8.0675	0.4471
WD13	PPFIA3	HLA-A24:02	QYRSYFMEL	10.9353	0.2811

WD13	PPP1R16B	HLA-A02:01	RLAPEQQMI	2.6027	0.2645
WD13	PRKD1	HLA-A02:01	NLSLDMWSV	12.7647	0.1657
WD13	PSEN2	HLA-A24:02	FFPAQKPAF	8.8337	0.1508
WD13	PSEN2	HLA-A24:02	KFFPAQKPAF	12.5068	0.2275
WD13	SETD3	HLA-A24:02	TYKTTIEEDKF	36.5	0.4478
WD13	TG	HLA-A24:02	QFHLDSKTF	2.9388	0.4483
WD13	TRIM14	HLA-A24:02	GELYHPVPLS	9.0082	0.2369
WD13	TRIM14	HLA-A24:02	LYHPVPLSFE	29.4483	0.2927
WD13	TRIM14	HLA-A24:02	YHPVPLSF	4.3308	0.2896
WD13	TRIM14	HLA-A24:02	LYHPVPLSF	3.0932	0.0046
WD13	TRIM14	HLA-A24:02	ELYHPVPLSF	2.4238	0.0619
WD13	ZNF331	HLA-A24:02	EYGKAFTRV	38.8333	0.3966
WD14	BGLAP	HLA-A02:01	YQWLGAPVL	5.5631	0.3009
WD14	CAMSAP3	HLA-A02:01	LLVSAVPMMA	2.8268	0.3634
WD14	DPPA5	HLA-A24:02	PYIEQVSKV	3.4932	0.2375
WD14	HAUS6	HLA-A24:02	KWSLSPLIKF	3.4139	0.2175
WD14	NBAS	HLA-A24:02	SFAEVLLRI	6.5325	0.2988
WD14	UBE2Q2	HLA-A24:02	GYVLGGGAL	8.0789	0.1017
WD14	UBE2Q2	HLA-A24:02	YVLGGGALF	16.9955	0.3494
WD14	ZNF780A	HLA-A24:02	QYQSIHAGI	4.5896	0.2327
WD22	ARHGAP15	HLA-A02:01	SELDGDLWT	6.2565	0.4039
WD22	ARHGAP15	HLA-A02:01	ELDGDWLTA	7.3187	0.4551
WD22	ASXL1	HLA-A02:01	AIGGGVARV	17.1963	0.3124
WD22	BCO2	HLA-A24:02	HFPKWLNFG	14.6989	0.0887
WD22	BIN1	HLA-A02:01	SLPDGSPAA	14.1682	0.2719
WD22	C1S	HLA-A02:01	SLIVEEFQV	15.3716	0.0615
WD22	CASP12	HLA-A24:02	IFQNGSTF	3.7896	0.4706
WD22	CEP295	HLA-A24:02	TSLPPPPTLF	23.2576	0.2711
WD22	CNTLN	HLA-A24:02	NYKEWMLYI	14.578	0.1878
WD22	DGKD	HLA-A02:01	GTWNDLARV	2.8351	0.3134
WD22	DSEL	HLA-A02:01	YIDIPETELV	7.7314	0.4917
WD22	ELAVL4	HLA-A24:02	AYGVKRLMF	14.3844	0.1365
WD22	EPB42	HLA-A02:01	SLIHSFNTRYL	12.3404	0.4843
WD22	EPG5	HLA-A24:02	EYSRIVQLF	5.659	0.0098
WD22	EPG5	HLA-A24:02	MEYSRIVQLF	12.339	0.1573
WD22	EPG5	HLA-A24:02	IMEYSRIVQLI	22.5263	0.3127
WD22	EPG5	HLA-A24:02	YSRIVQLF	42.4286	0.4461
WD22	FBXO21	HLA-A24:02	YYAKKILYYLV	4.3754	0.3357
WD22	FILIP1	HLA-A02:01	KLMNKEDQL	3.4354	0.2383
WD22	FLRT2	HLA-A24:02	SYTPPTPTTL	7.9598	0.1052
WD22	GABRB3	HLA-A02:01	YLMGGFVFV	2.4275	0.2546
WD22	GATA2	HLA-A02:01	GSHLFGFPPT	4.3483	0.2824
WD22	GATA2	HLA-A02:01	SHLFGFPPTL	2.5017	0.1186
WD22	GLTSCR1L	HLA-A02:01	FLEDELESSL	8.4919	0.1736
WD22	GPR142	HLA-A02:01	GLTTLFTLL	2.4324	0.4735
WD22	GRINA	HLA-A24:02	PWGPLPTAPI	6.8775	0.2617

WD22	ICOSLG	HLA-A24:02	LWGDFSLRLI	4.6253	0.2883
WD22	ICOSLG	HLA-A24:02	LMSPAGMLV	10.1154	0.311
WD22	ICOSLG	HLA-A24:02	LWGDFSLRL	5.6365	0.3448
WD22	ICOSLG	HLA-A02:01	MLWGDFSL	3.7113	0.4378
WD22	KCNG4	HLA-A24:02	SYWWAIIF	11.4478	0.413
WD22	KIR2DL4	HLA-A24:02	MFMSPTVII	3.9979	0.3775
WD22	KLC1	HLA-A24:02	ILPKSLWMTF	2.9636	0.2954
WD22	LAMB4	HLA-A24:02	SWWAGTRS	9.9461	0.3028
WD22	LAMB4	HLA-A24:02	SWWAGTRS	6.5045	0.3834
WD22	LAMB4	HLA-A24:02	VIPPLVISW	2.2563	0.2903
WD22	MAP7D1	HLA-A02:01	GMPDLLEGA	9.0913	0.4232
WD22	METTL23	HLA-A24:02	QYLWFHRS�	12.0805	0.4312
WD22	MROH7	HLA-A24:02	EYAKNSRAF	20.8696	0.2646
WD22	MYBPC2	HLA-A02:01	LLDTSRVHV	3.8701	0.3652
WD22	MYBPC2	HLA-A24:02	ITPEIWELL	2.2925	0.4936
WD22	MYO3A	HLA-A02:01	LLTDIVLLL	8.4315	0.0438
WD22	NRXN1	HLA-A02:01	GLFVGGLPL	4.5003	0.2637
WD22	OR13C3	HLA-A24:02	IFFSGCAVQM	3.3432	0.3434
WD22	OR1S1	HLA-A24:02	MWPRFGILL	2.1333	0.1381
WD22	OR5K3	HLA-A02:01	VLFEVFFAI	2.171	0.1317
WD22	PPP3CB	HLA-A02:01	FLQNNNLLL	2.4797	0.0522
WD22	PSEN2	HLA-A02:01	YMPFTEDTPS	3.9421	0.3011
WD22	PTPRC	HLA-A24:02	IYDLHKKRF	15.9281	0.1797
WD22	PXDN	HLA-A24:02	QYITYQHWL	4.9856	0.1768
WD22	RERG	HLA-A02:01	MQWGEGFV	3.979	0.4419
WD22	RPH3A	HLA-A24:02	PFSVDGFSF	4.6832	0.1576
WD22	SCML2	HLA-A02:01	SLHEVKFQM	17.5444	0.2125
WD22	SERPINI2	HLA-A24:02	IFIKRFPYL	14.2285	0.1215
WD22	SERPINI2	HLA-A24:02	KIPNLQWIF	22.3421	0.2322
WD22	SHE	HLA-A24:02	DYADPYDAK	12.6318	0.3464
WD22	SLC9C2	HLA-A02:01	SIASIFLEI	36.8947	0.3576
WD22	SPATA31E1	HLA-A24:02	AFGPPEPLL	5.8279	0.4606
WD22	TACC2	HLA-A02:01	SLQKEQLRV	2.0681	0.4954
WD22	TAS2R38	HLA-A24:02	LWMIANQAI	2.1402	0.3818
WD22	TENM4	HLA-A24:02	LWYDIRDRI	5.67	0.3622
WD22	TERT	HLA-A02:01	ALLDGARGA	11.0134	0.4662
WD22	TKTL2	HLA-A02:01	KLPVEDSPQI	16.3194	0.341
WD22	TLX3	HLA-A24:02	KYLASAERAA	2.0317	0.455
WD22	TPTE	HLA-A02:01	HLYNWNLPL	4.2723	0.2194
WD22	TRAJ39	HLA-A24:02	NNYAGNML	5.6694	0.0655
WD22	TRAJ39	HLA-A24:02	YAGNMLTF	5.7988	0.2586
WD22	TRAJ39	HLA-A24:02	NYAGNMLTF	20.0348	0.2946
WD22	TSSC1	HLA-A24:02	SYDGRLVINV	8.1138	0.2007
WD22	ZNF214	HLA-A24:02	KYKNFMPWC	12.3527	0.1003

LAQ3:

1)

distinct binding affinity thresholds were applied. The distribution of these identified peptides across the four samples, categorized by both general (Rank < 2%) and high-stringency (Rank < 0.5%) criteria, is summarized in the table below:

Strong Neoantigens	tumour Neoantigens	Sample
66	561	WD06
37	360	WD13
8	64	WD14
70	483	WD22

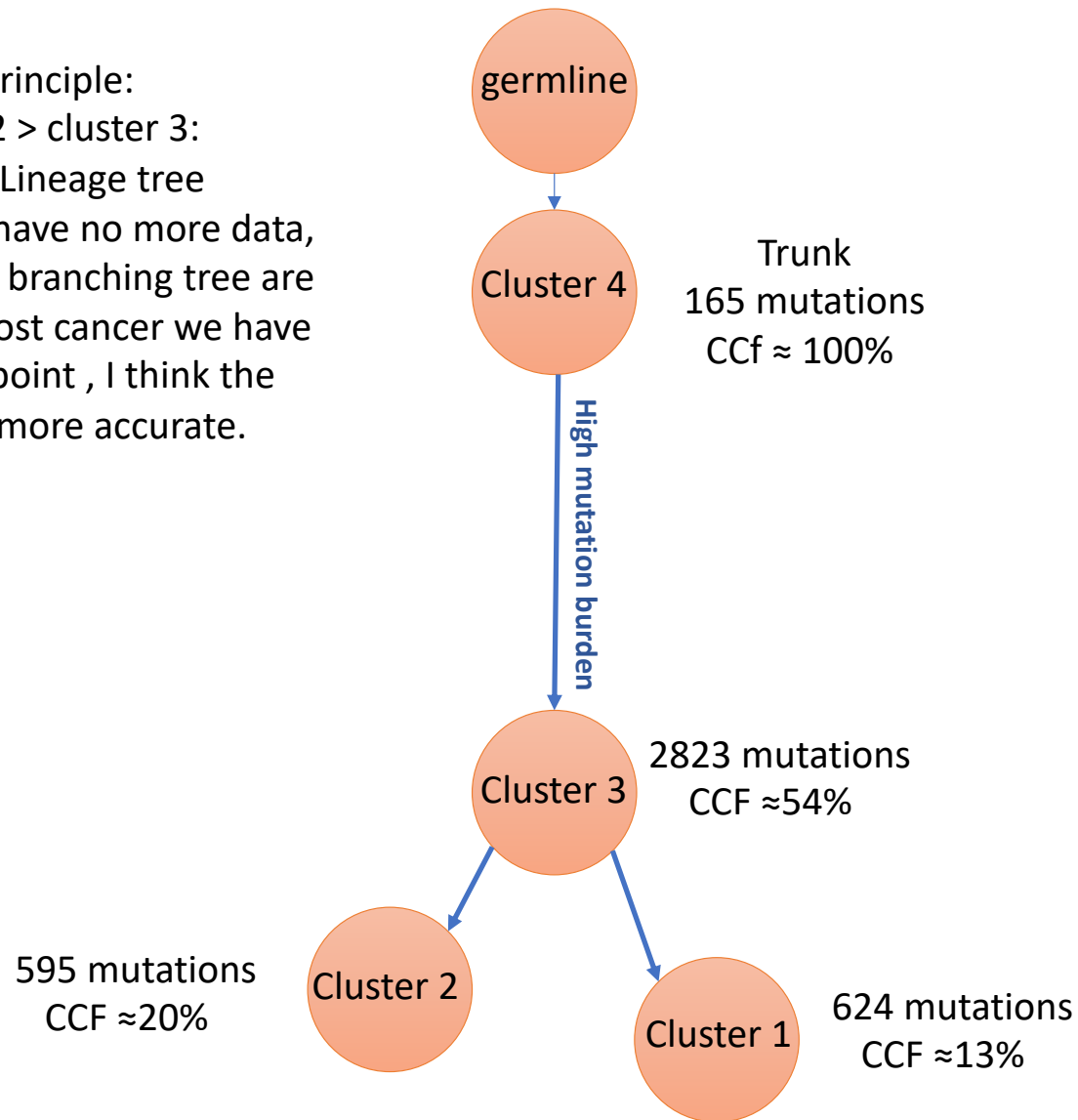
2) file name : we student could not submit csv , so I change the file to pdf and add it in this file

3) Our analysis revealed a complete absence of shared tumor neoantigens, indicating that neoantigenic profiles are highly patient-specific. From a therapeutic perspective, this necessitates a customized approach for each patients. Personalized interventions, such as patient-specific neoantigen vaccines, appear to be a more viable path for achieving optimal clinical outcomes given the private nature of these mutations.

Question3:LU_4

Based on pigeonhole principle:

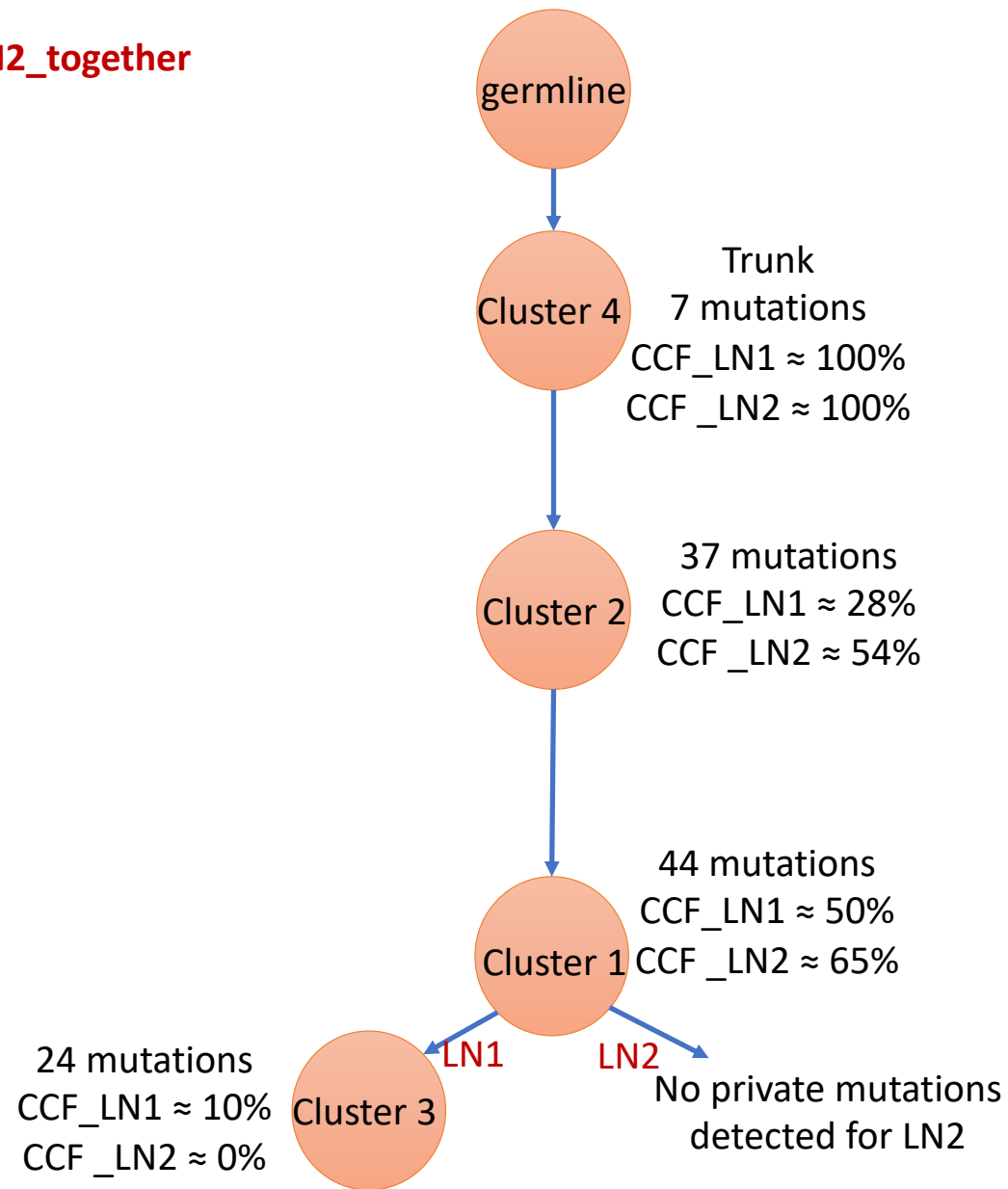
- If cluster1 + cluster2 > cluster 3:
It should be Lineage tree
- In this example we have no more data, so both lineage and branching tree are acceptable. As in most cancer we have branching at some point, I think the branching model is more accurate.



Question3:LU-4

Cluster	size	Mean Vaf	Scaled Vaf	CCF %	Status	Zygotity
4	165	0.447	1.0000000	100	Founder / Trunk Clone	Heterozygous
3	2823	0.242	0.5413870	54.13870	Major Clone (Dominant Subclone)	Heterozygous
2	595	0.089	0.1991051	19.91051	Subclone	Heterozygous
1	624	0.058	0.1297539	12.97539	Subclone	Heterozygous

Question6:SP4_LN1_LN2_together



Question6:SP4_LN1_LN2_together

Cluster	size	Mean Vaf LN1%	Mean Vaf LN2%	CCF LN1 %	CCF LN2 %	Zygotity	Status
1	44	40.420419	30	49.9	65	Heterozygous	Shared Subclone
2	37	23.174340	25	28.65393	54	Heterozygous	Shared Subclone
3	24	8.374064	0.16	10.35412	0.35	Heterozygous	Private Subclone (LN1)
4	7	80.876643	45.9	100	100	Homozygous (LN1)/ Heterozygous (LN2) IN LN1 : It might be a result of LOH or Copy number gain	Trunk

SAQ 1:

Briefly describe the characteristics of the cancer genome

Cancer develops gradually through the accumulation of genetic mutations over time. If a mutation provides a cell with a survival or growth advantage, that cell expands, and the tumor evolves through a process remarkably similar to natural selection.

Genomic mutations in cancer occur across multiple scales. At the smallest level, we see single-nucleotide substitutions (SBS) which reflect specific biological processes like DNA damage or defects in repair pathways. However, cancer alterations aren't limited to this scale. Many tumors undergo massive structural rearrangements, changes in gene copy numbers, the deletion of tumor suppressor genes, or the amplification of oncogenes. Entire chromosomes can even be lost or gained, a state known as aneuploidy.

These large-scale disruptions fuel genomic instability. In extreme cases, cancer cells may undergo whole-genome doubling (WGD). Though it seems counterintuitive, this can actually have a protective effect, increasing the cell's tolerance for subsequent instabilities.

Ultimately, tumors are not genetically uniform. Even within a single tumor, different cells can have different mutations. This genetic heterogeneity is the result of continuous mutation and selection, allowing the tumor to adapt to environmental pressures and clinical treatments. In total, cancer progression is a dynamic, evolutionary journey rather than a static state.

SAQ2:

1. What is cancer immunotherapy? (2 marks)

Cancer immunotherapy is a treatment approach designed to activate the body's immune system or restore its lost capacity to better identify and destroy cancer cells. Unlike chemotherapy, which directly targets fast-growing cells, immunotherapy focuses on strengthening the body's own immune response against the disease. Essentially, it aims to overcome the mechanisms tumors develop to suppress the immune system, creating a more effective and sustainable anti-tumor response.

2. Give two examples of cancer immunotherapy, and also brief describe the underlying mechanism of each immunotherapy? (4 marks)

Two key examples of cancer immunotherapy include immune checkpoint inhibition and therapies targeting tumor-associated macrophages (TAMs).

In immune checkpoint inhibition, killer T cells (CD8) are often already present within the tumor microenvironment, but their activity is suppressed by inhibitory signals. While these cells are naturally capable of destroying cancer, their presence is usually a sign of a better prognosis only if they can be "unlocked." Checkpoint inhibitors block these inhibitory signals, reactivating the T cells to strengthen the anti-tumor response. This approach is particularly effective in "inflamed" tumors that already contain a high density of T cells.

The second approach targets tumor-associated macrophages (TAMs). These cells are abundant in many cancers and are typically linked to a poor prognosis because they promote tumor growth, angiogenesis, and immunosuppression. Consequently, certain therapies aim to either deplete the number of TAMs or reprogram them into an anti-tumor state to bolster the body's overall immune response.

3. Briefly describe strategies that tumour cells take to evade immune surveillance. (4 marks)

Tumor cells employ various strategies to evade immune detection, with one of the most critical being the creation of a suppressive immune microenvironment. By recruiting regulatory T cells and myeloid-derived suppressor cells, tumors actively dampen the activity of killer T cells and weaken the overall immune response. Additionally, dendritic cells can become dysfunctional, losing their ability to effectively prime and activate T cells.

In many cases, even when T cells are present, they are physically blocked from infiltrating the tumor core due to alterations in the extracellular matrix and stromal structure—a state known as an "immune-excluded" phenotype. Furthermore, tumors can further hinder immune function by inducing immune tolerance in lymph nodes and developing abnormal blood vessels that make effective immune cell penetration nearly impossible.